

XV Ibero-American Interuniversity Mathematics Competition - Rio de Janeiro, Brazil
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by YLG_123

- 1 Determine all the pairs of positive real numbers (a, b) with $a < b$ such that the following series

$$\sum_{k=1}^{\infty} \int_a^b \{x\}^k dx = \int_a^b \{x\} dx + \int_a^b \{x\}^2 dx + \int_a^b \{x\}^3 dx + \dots$$

is convergent and determine its value in function of a and b .

Note: $\{x\} = x - \lfloor x \rfloor$ denotes the fractional part of x .

- 2 A toymaker has k dice at his disposal, each with 6 blank sides. On each side of each of these dice, the toymaker must draw one of the digits $0, 1, 2, \dots, 9$.

Determine (in terms of k) the largest integer n such that the toymaker can draw digits on the k dice such that, for any positive integer $r \leq n$, it is possible to choose some of the k dice and form with them the decimal representation of r .

Note: The digits 6 and 9 are distinguishable: they appear as 6 and 9.

- 3 Given a 3×3 symmetric real matrix A , we define $f(A)$ as a 3×3 matrix with the same eigenvectors of A such that if A has eigenvalues a, b, c , then $f(A)$ has eigenvalues $b+c, c+a, a+b$ (in that order). We define a sequence of symmetric real 3×3 matrices $A_0, A_1, A_2, \dots$ such that $A_{n+1} = f(A_n)$ for $n \geq 0$. If the matrix A_0 has no zero entries, determine the maximum number of indices $j \geq 0$ for which the matrix A_j has any null entries.

- 4 For a positive integer n , $\sigma(n)$ denotes the sum of the positive divisors of n . Determine

$$\limsup_{n \rightarrow \infty} \frac{\sigma(n^{2023})}{(\sigma(n))^{2023}}$$

Note: Given a sequence (a_n) of real numbers, we say that $\limsup_{n \rightarrow \infty} a_n = +\infty$ if (a_n) is not upper bounded, and, otherwise, $\limsup_{n \rightarrow \infty} a_n$ is the smallest constant C such that, for every real $K > C$, there is a positive integer N with $a_n < K$ for every $n > N$.

- 5 Given a positive integer $k > 1$, find all positive integers n such that the polynomial

$$P(z) = z^n + \sum_{j=0}^{2^k-2} z^j = 1 + z + z^2 + \dots + z^{2^k-2} + z^n$$

has a complex root w such that $|w| = 1$.

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- 6** Let n be a positive integer. We define $f(n)$ as the number of finite sequences $(a_1, a_2, \dots, a_k)$ of positive integers such that $a_1 < a_2 < a_3 < \dots < a_k$ and

$$a_1 + a_2^2 + a_3^3 + \dots + a_k^k \leq n.$$

Determine the positive constants α and C such that

$$\lim_{n \rightarrow \infty} \frac{f(n)}{n^\alpha} = C.$$
